BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

Section-A (Multiple Choice Questions (MCQ's))

Note: Q.1 Choose the correct answer for each from the given options:

Q1. The ratio of out put to input is called:

(a) Energy
(b) Work
(c) Power
(d) Efficiency

Q2. If a machine performs 20J of work is 10sec, then its power is:

(a) 200 watt
(b) 20 watt
(c) 2 watt
(d) 0.2 watt

Q3. A gas in a container of fixed volume is heated what happen to the molecules of the gas.

(a) They collide less frequently
(b) They expand
(c) They move faster
(d) They move further apart

Q4. Law of conservation of momentum define that the total momentum of a system of two bodies before and after collision.

(a) remains constant
(b) retain more momentum
(c) loses some momentum
(d) None of above is true

Q5. Stability of a racing car is increased by:

(a) increasing its height
(b) raising its centre of gravity
(c) decreasing its width
(d) lowering its centre of gravity

Q6. If a metal block applies a force of 20N on an area of 5cm2. Find the Pressure being applied by the block on the area:

(a) 100 N cm2
(b) 0.8 Ncm2
(c) 0.25 Ncm2
(d) 4 Ncm2

Q7. Heat is the form of:

(a) Pressure
(b) Weight
(c) Energy
(d) All of these

Q8. The melting point of ice at normal atmospheric pressure is:

(a) 0C
(b) 0 K
(c) 100C
(d) 70F

Q9. A fixed temperature at which a pure liquid boils is called:

(a) melting point
(b) freezing point
(c) boiling point
(d) both (a) and (b)

Q10. Numerical value of G is:

(a) G = 6.673 x 10-11 Nm2kg-2
(a) G = 6,673 x10-11Nm2 kg2
(c) G = 6.763 x 10-11N2 kg2
(d) G = 6.763 x1011 Nm2 kg2

Q11. A communication satellite completes its one revolution.

(a) 6
(b) 12
(c) 18
(d) 24

Q12. A burning candle is an example of state of matter,

(a) Gas
(b) Liquid
(c) Solid
(d) All of these

(SECTION - B) (Short answers)

> Note: Answer any EIGHT of the following questions. Each questions carries 03 marks:

Q1. Why a write fence is designed in the helmet of batsman and How does it prevent form injuries?

Q2. The mass of an object is 60kg. Find its weight on: (i) Earth (ii) Moon (iii) Mars assume the acceleration due to gravity on Earth- 9.8m.s-2 on moo=1.6 m.s2 and mars-3.7m.s

Q3. How will you determine the centre of mass or center of gravity.

Q4. How anti-loc braking system prevents the risk of sliding?

Q5. Calculate the spring constant for a spring which extends by a distance of 3.5cm when a [illegible in source scan] of 14 N [illegible in source scan] is hung from its end.

Q6. Define the terms: (i) Critical Velocity (ii) Communication Satellite

Q7. The pressure on 9cm3 of oxygen gas is doubled at a fixed temperature what will its volume be3come?

Q8. Why the kinetic model of matter is called kinetic?

Q9. Define work? Drive the equation work + F.D Cos(.)

Q10. Write notes on Tidal energy and Geothermal energy.

Q11. Differentiate between heat and temperature.

Q12. Explain specific heat capacity.

SECTION- C (Descriptive Answers)

> Note: Answer any FOUR of the following questions. Each question carries 06 marks:

Q13. Why solids increase in size on heating? Explain

Q14. Define solar energy and its importance in Pakistan?

Q15. What will be the Kinetic energy of abody of mass 50 kg driving a bike with velocity of 2m.s?

Q16. Why weight ofg an object does not remain same eveyr where on eEarth?

Q17. Define Newton's section Law of motion.,

Q18. Write notes on any TWO of the following: (a) Brownian motion (b) Efficiency (c) Importancr of Physis